

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA5112

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5, BL6)

Assessment Tool Weightage: 10%

QUESTIONS: 20

TOTAL MARKS: 20

Annexure A
MOCK INTERVIEW

Code of Conduct:

I E. Lakshmi Narayanan(192524249) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E. Lakshmi Narayanan

1. Which would you prioritize in a secure network design: performance or security, and how would you balance both?

Answer:

I would prioritize security because a fast network is not useful if it is vulnerable to attacks. However, security should not significantly reduce performance. I would balance both by using optimized firewalls, efficient encryption, network segmentation, load balancing, and proper monitoring.

2. Difference between stateful and stateless firewalls. Which is suitable for enterprise networks?

Answer:

A **stateless firewall** checks each packet independently based on rules such as IP address and port number. A **stateful firewall** tracks active connections and understands whether a packet belongs to an existing session. Stateful firewalls are generally more suitable for enterprise networks because they provide better context-aware security.

3. If an IDS generates too many false positives, how would you optimize it?

Answer:

I would first analyze the alerts and identify common sources of false positives. Then I would tune detection rules, adjust thresholds, remove unnecessary signatures, and whitelist trusted activities where appropriate. I would also use machine learning or behavioral analysis to improve detection accuracy.

4. How would you design a real-time IDS for high-speed networks?

Answer:

I would use high-performance sensors, efficient feature extraction, parallel processing, and optimized detection algorithms. I would prioritize important traffic and use distributed IDS sensors. Alerts would be sent to a centralized monitoring system such as a SIEM for real-time analysis.

5. What are the challenges in detecting zero-day malware?

Answer:

Zero-day malware is difficult to detect because there is no previous signature or known pattern. Attackers can also modify malware to avoid traditional detection. I would address this using behavioral analysis, sandboxing, anomaly detection, machine learning, and endpoint detection and response.

6. Difference between signature-based and anomaly-based IDS?

Answer:

Signature-based IDS compares network activity against known attack signatures. It is very effective for known attacks but has difficulty detecting new attacks. **Anomaly-based IDS** learns normal behavior and detects unusual activity. I would use signature-based detection for known threats and anomaly-based detection for unknown or emerging threats. Using both provides better coverage.

7. Network under DDoS attack – immediate and long-term mitigation?

Answer:

Immediately, I would identify the attack traffic, apply rate limiting, block malicious sources where practical, and use upstream DDoS protection or traffic scrubbing. For long-term protection, I would use a DDoS protection service, scalable infrastructure, redundant networks, traffic monitoring, and an incident response plan.

8. How would you secure remote access to an enterprise network?

Answer:

I would use a secure VPN with strong encryption and **multi-factor authentication**. I would also implement role-based access, least-privilege policies, endpoint security, device verification, and continuous monitoring. This ensures that only authorized users and devices can access corporate resources.

9. Limitations of traditional antivirus software?

Answer:

Traditional antivirus mainly depends on known malware signatures, so it may not detect new or zero-day threats. Modern solutions overcome this using behavioral detection, machine learning, cloud-based threat intelligence, sandboxing, and endpoint detection and response.

10. How would you design a secure email filtering system?

Answer:

I would use multiple layers such as spam filtering, malware scanning, URL analysis, attachment scanning, sender authentication, and phishing detection. Technologies such as **SPF, DKIM, and DMARC** can help verify email authenticity. Suspicious messages can be quarantined for further inspection.

11. If an attacker bypasses firewall rules, what additional security layers would you implement?

Answer:

I would use **defense in depth**. Additional layers would include IDS/IPS, endpoint protection, network segmentation, access control, MFA, encryption, SIEM monitoring, vulnerability management, and regular security audits. This ensures that bypassing one security layer does not compromise the entire network.

12. How would you evaluate network segmentation?

Answer:

I would check whether sensitive systems are isolated from normal user networks and whether unauthorized communication between segments is blocked. I would perform penetration testing, review firewall rules, monitor traffic between VLANs, and measure whether an attack in one segment can spread to another.

13. What role does encryption play in preventing malware attacks?

Answer:

Encryption does not directly prevent malware from infecting a system. Its main purpose is to protect data from unauthorized access. However, encryption protects sensitive information during transmission and storage, reducing the impact of data theft. Strong authentication and endpoint security are still required to prevent malware.

14. How would you secure a cloud-based network infrastructure?

Answer:

I would use strong identity and access management, MFA, least-privilege permissions, encryption, network segmentation, secure APIs, firewalls, continuous monitoring, and regular vulnerability assessments. Cloud resources should also follow a **Zero Trust** approach where access is continuously verified.

15. Importance of log analysis and monitoring?

Answer:

Logs provide information about network activity, user actions, system errors, and security events. Continuous monitoring helps identify suspicious behavior and security incidents quickly. Log analysis also supports incident investigation, compliance, and forensic analysis.

16. How would you respond to a security breach?

Answer:

I would follow the incident response process:

Identify → Contain → Eradicate → Recover → Review

First, I would identify the affected systems and contain the attack. Then I would remove the threat, restore systems safely, investigate the root cause, and improve security controls to prevent similar incidents.

17. Advantages and limitations of NGFW?

Answer:

A **Next-Generation Firewall (NGFW)** provides advanced features such as application awareness, user-based policies, intrusion prevention, and deep packet inspection.

Its advantages include better visibility and stronger security. Its limitations include higher cost, increased configuration complexity, and possible performance overhead from advanced inspection.

18. How would you integrate IDS/IPS with SIEM?

Answer:

I would send IDS/IPS alerts and other security logs to the SIEM. The SIEM can correlate events from multiple sources, identify attack patterns, prioritize alerts, and generate centralized dashboards and notifications. This improves visibility and helps security teams respond faster.

19. Best practices for patch management and vulnerability assessment?

Answer:

First, I would regularly scan systems for vulnerabilities and classify them based on risk. Critical vulnerabilities should be patched as quickly as possible. I would test patches before deployment, maintain an asset inventory, automate patch deployment where appropriate, and continuously verify that patches were successfully applied.

20. How would you design a future-proof network security architecture?

Answer:

I would use a **defense-in-depth and Zero Trust architecture**. It would include secure identity management, MFA, network segmentation, encryption, NGFW, IDS/IPS, endpoint security, SIEM, automated threat detection, vulnerability management, and regular security testing. The architecture should also be scalable and adaptable so that new technologies such as **AI-based detection, cloud security, and post-quantum cryptography** can be incorporated as threats evolve.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism